

Antika Dey

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SUMMARY

Senior Data Engineer with 6+ years of experience designing and scaling cloud-native data platforms on AWS and Databricks. Expertise in migrating legacy, on-prem databases and workflows to modern cloud-native architectures, including a full on-prem warehouse migration to AWS that cut infrastructure costs by 50%. Specializes in PySpark and dbt-based ETL/ELT pipelines, real-time streaming with Kafka and Kinesis, and Lakehouse formats (Iceberg, Delta Lake) supporting petabyte-scale analytics. Skilled at orchestrating production workflows with Airflow, optimizing Redshift/Athena performance, and delivering BI reporting that drives faster business decisions.

TECHNICAL SKILLS

Languages: Python, Spark SQL, PySpark, Scala, Bash

Data Engineering & Streaming: Apache Spark, Apache Airflow, dbt/dbt-databricks, Apache Kafka, Kinesis Data Streams, Delta Live Tables, Informatica PowerCenter, ETL/ELT Design

Lakehouse & Storage: Apache Iceberg, Delta Lake, Parquet/JSON/CSV, Unity Catalog, ACID Transactions, Schema Evolution, CDC

Databases: PostgreSQL, MySQL, DynamoDB, Amazon Redshift, pgvector, Star/Snowflake Schema Design

Cloud & Platforms: AWS (S3, Redshift, Glue, EMR, Athena, DynamoDB, Kinesis, CloudWatch, CloudTrail, IAM), Databricks

CI/CD & Orchestration: Autosys, Jenkins, Apache Airflow, Git, GitHub Actions, Bitbucket, Docker, Docker Compose

BI, Quality & Observability: Power BI, Qlik Sense, Databricks Lakeview, Streamlit, Great Expectations, dbt Tests, CloudWatch/CloudTrail, IAM Governance

AI/ML & Automation: RAG, sentence-transformers, Ollama, Vector Search

Concepts: Data Lakehouse Architecture, Distributed Systems, SLA Management, Data Governance, Data Lineage

Certifications: AWS Certified Cloud Practitioner – Associate, Claude Code 101, AI Fluency, Claude 101

PROFESSIONAL EXPERIENCE

Cognizant Technology Solutions | (Client – Prudential Financial) – Plano, TX

Oct 2024 - Present

Senior Data Engineer

- Migrated an on-prem data warehouse to AWS (S3 + Redshift), cutting infrastructure costs by 50% and improving query performance by 80%, enabling faster analytics for business stakeholders.
- Designed and built scalable PySpark ETL pipelines on Databricks and AWS Glue/EMR to ingest structured and semi-structured data (Parquet, JSON, CSV) into a cloud Lakehouse, improving pipeline reliability and throughput.
- Architected a production Lakehouse table layer using Apache Iceberg with ACID transactions and schema evolution, enabling incremental processing at petabyte scale.
- Built and managed end-to-end orchestration workflows using Apache Airflow (DAG authoring, SLA alerting, retry logic), replacing legacy Autosys jobs and reducing pipeline failures by 35%.
- Engineered a dbt transformation layer with modular SQL models, macros, tests, and documentation, enabling 4 downstream analytics teams to self-serve data quality checks and lineage visibility.
- Developed real-time ingestion pipelines using Apache Kafka and Kinesis Data Streams for event-driven architectures, reducing data latency from hours to under 5 minutes for critical business events.
- Implemented S3 data lake best practices (partitioning, incremental loads, optimized file formats), reducing query latency and lowering storage/query costs by 25%.
- Optimized Amazon Redshift performance (distribution styles, sort/dist keys, query tuning), supporting high-concurrency analytics workloads with zero SLA breaches.
- Enabled serverless analytics using Amazon Athena with optimized Parquet/Iceberg tables, eliminating dedicated cluster overhead for ad-hoc query workloads.
- Hardened platform observability via CloudWatch dashboards, CloudTrail audit logs, and IAM policy governance, reducing mean time to incident resolution (MTTR) by 40%.
- Delivered Power BI dashboards with incremental refresh, reducing refresh times and enabling near real-time reporting for business stakeholders.
- Partnered with business analysts and data scientists to define data contracts and SLAs for critical financial reporting pipelines, reducing downstream data quality escalations by 30%.
- Automated data lineage and impact-analysis reporting using dbt's metadata layer, cutting the time to assess schema-change risk from days to hours for cross-functional teams.

Associate Data Engineer

- Designed and delivered end-to-end ETL pipelines on AWS (Glue, EMR, PySpark) processing 500M+ daily records across structured and semi-structured sources with full SLA compliance.
- Introduced dbt for the SQL transformation layer with modular models, schema and data tests, and automated documentation, reducing ad-hoc data fixes by 45% and improving analyst trust in data.
- Migrated legacy Informatica PowerCenter workflows to cloud-native PySpark + Airflow pipelines, leveraging pushdown optimization and partitioning to cut runtime by 30% and failures by 40%.
- Built modular SQL transformation layers and reusable stored procedures for analytics use cases, applying dbt-equivalent patterns for data modeling, testing, and pipeline maintainability.
- Implemented Delta Lake on S3 for ACID-compliant data storage with upserts and merge operations, enabling CDC (Change Data Capture) patterns for 15 high-velocity data feeds.
- Designed star and snowflake dimensional models on Redshift supporting BI reporting for Power BI and Qlik Sense dashboards used by 150+ business users daily.

Hewlett Packard Enterprise – West Bengal, India**Jun 2018 - Jul 2019***Software Engineer*

- Developed backend data processing features using Python and SQL, improving system stability and data reliability.
- Optimized complex SQL queries and joins across tables, reducing query runtimes by identifying and indexing key columns.
- Analyzed and refactored legacy stored procedures to remove redundant logic, improving execution efficiency and code maintainability for a 50K-line codebase.
- Designed data flows and identified source-to-target mappings, improving data lineage and dependency tracking.
- Implemented logging, validation, and exception handling to improve system robustness and reduce production issues.

PROJECTS

Tapstream – Real-Time E-Commerce Clickstream Platform (Kafka, Databricks DLT, dbt, Airflow, Streamlit)

- Built an end-to-end streaming data platform simulating live e-commerce events (page views, add-to-cart, purchases) and processing them through a bronze/silver/gold Delta Live Tables pipeline on Databricks.
- Tunneled a locally run Kafka cluster to Databricks via ngrok, configuring multi-listener networking so the DLT pipeline could consume events continuously in real time.
- Built dbt-databricks models on top of the Delta Lakehouse for daily revenue, conversion funnel, and cohort retention analytics, and orchestrated syncs, dbt builds, and data-quality checks with Airflow.
- Implemented a Postgres serving layer synced from gold-layer aggregates and a Streamlit dashboard with live and historical views, plus Great Expectations checks for freshness and data quality.

SchemaMind – AI-Powered Data Catalog Assistant (dbt, pgvector, Airflow, Ollama, Streamlit)

- Built a fully local, zero-API-cost RAG assistant that answers natural-language questions about a dbt/warehouse schema using retrieved context instead of hallucinated answers.
- Designed a metadata pipeline that extracts dbt manifest and information_schema data, embeds it locally with sentence-transformers, and stores vectors in Postgres/pgvector, avoiding a separate vector database.
- Integrated a locally hosted LLM (Ollama/Llama 3.2) for answer generation and built a Streamlit chat UI, with an Airflow DAG to nightly refresh dbt docs and re-embed metadata so answers stay current.

US Emissions Breakdown Analysis (Databricks SQL, Lakeview Dashboards)

- Authored a series of Databricks SQL notebooks to explore and break down U.S. emissions data by source and region.
- Built an interactive Lakeview dashboard on Databricks to visualize emissions trends and support ad-hoc analytical queries.
- Wrote parameterized SQL queries to break down emissions by sector, state, and year-over-year trend, enabling faster identification of the highest-impact emission sources.

EDUCATION

Florida State University – Tallahassee, Florida**Jul 2019 – Apr 2021**

MS in Computer Science

Maulana Abul Kalam Azad University of Technology – West Bengal, India**Jul 2014 – May 2018**

BTech in Computer Science and Engineering